Complex Analysis Homework #11

1. A *circular polygon* $\mathcal{C}$ is a simply connected open set of $\mathbb{C}$ whose boundary is a union of circular arcs and is homeomorphic to a circle. Show that any Riemann map $\Delta \to \mathcal{C}$ extends to a homeomorphism $\overline{\Delta} \to \overline{\mathcal{C}}$.

Suppose for contradiction that a Riemann map $\Delta \to \mathcal{C}$ could not be extended continuously to the boundary. That is, we would have two sequences in Δ converging to some point on the boundary, $z_n, z_n' \to z_0 \in S^1$, such that their images converged to distinct points on $\partial\mathcal{C}$. In other words, there is some positive δ such that $|f(z_n) - f(z_m')| > \delta$ for all n, m.

Consider the following series of calculation. Consider circles α_r of radius r about z_0. For sufficiently large n, m and sufficiently small r there are points in both sequences which lie on α_r. Now, by the fundamental theorem of calculus we have

$$\begin{aligned}
\delta < \left|f(z_n) - f(z_m')\right| &= \left|\int_{\alpha_r} f'(\xi)d\xi\right| \\
&= \left|\int_{\theta_1(r)}^{\theta_2(r)} f'(z_0 + re^{i\theta})rd\theta\right| \\
&\leq \left|\int_{\theta_1(r)}^{\theta_2(r)}\right| \left|f'(z_0 + re^{i\theta})\right| rd\theta
\end{aligned}$$

where the second equality is given by parameterizing α_r and the final inequality given by the triangle inequality; the angles $\theta_1(r), \theta_2(r)$ are the angles on which z_n, z_m' respectively lie on α_r. Now, squaring both sides and then applying Cauchy-Schwarz (taking $h = fr^{1/2}$, $g = r^{1/2}$) gives

$$\begin{aligned}
\delta^2 &\leq \int \left|f'(z)\right|^2 rd\theta \int_{\theta_1(r)}^{\theta_2(r)} rd\theta \\
&\leq \int \left|f'(z)\right|^2 rd\theta \int_0^{2\pi} rd\theta \\
&\leq \int \left|f'(z)\right|^2 rd\theta \, (2\pi r)\,.
\end{aligned}$$

Rearranging this inequality and then integrating over r gives

$$\int_{r=0}^{R} \frac{\delta^2}{r} dr \leq \int_r \int_\theta \left|f'(z)\right|^2 rd\theta dr = \int\int \left|f'\right|^2 dA = \text{Area}(f(\Delta)),$$

where the last equality is given by the fact that the Jacobian for the appropriate transformation is $|f'(z)|^2$. Now finally, notice that the left hand side diverges, but the right hand side is finite, since it is the area of a bounded open set. A contradiction, and so it must have been that f extended continuously to the boundary.

As a final note, notice that the hypothesis that the boundary be homeomorphic to the circle means that our boundary cannot have any self intersections. I.e. we have a "nice boundary", and the situation really is like the polygon case.

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2. Let $f : \Delta \to \Delta$ be a holomorphic function that extends continuously to $\overline{\Delta} \to \overline{\Delta}$, with $f(S^1) \subset S^1$. Show that f can be extended to a holomorphic function on $\mathbb{C}$ with finitely many points removed.

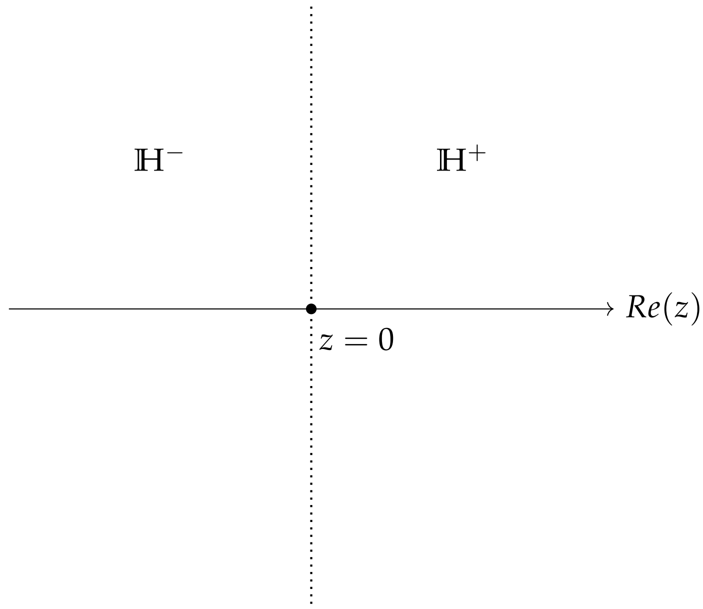
Recall that we have a biholomorphism from the $\Delta \to \mathbb{H}$ given by

$$h(z) = i \cdot \frac{1+z}{1-z}.$$

This map extends continuously to the boudaries $\bar{\Delta} \to \overline{\mathbb{H}} \cup \infty$ with $h(1) = \infty$. And so we can define a holomorphic function $g : \mathbb{H} \to \mathbb{H}$ by $g = h \circ f \circ h^{-1}$ which extends continuously to the actual boundary $\bar{\mathbb{H}} \cup \infty$ and $g(\bar{\mathbb{H}} \cup \infty) \subseteq \bar{\mathbb{H}} \cup \infty$. The preimages of 1 under f are those which get sent to infinity under g.

Suppose for a moment that we have a single point on the real line which gets sent to infinity. Say that we translate our coordinates so that this point is $z = 0$. Let us denote the "right half upper plane" by $\mathbb{H}^+ := \{z \in \mathbb{H} : Re(z) > 0\}$ and similarly let $\mathbb{H}^-$ denote the "left upper half plane".

Now, the restriction of g to $\mathbb{H}^+$ defines a holomorphic function which extends continuously to the interval $(0, \infty)$ with the image of $(0, \infty) \subseteq \mathbb{R}$. The Schwarz reflection principle then allows us to extend $g|_{\mathbb{H}^+}$ to a holomorphic function on the entire right half plane $\mathbb{R}^+ := \{z \in \mathbb{C} : Re(z) > 0\}$. The same argument means that we can extend $g|_{\mathbb{H}^-}$ to a holomorphic function on the entire left half plane. We now have a holomorphic function $\mathbb{C} \setminus L \to \mathbb{C}$ where L is the vertical line through $z = 0$. Let us denote this function by $\tilde{g}$.



We can extend $\tilde{g}$ further to a function $G : \mathbb{C} \setminus \{z = 0\} \to \mathbb{C}$ in the following way

$$G(z) = \begin{cases} \tilde{g}(z) & z \notin L \\ g(z) & z \in L \cap \mathbb{H} \\ \overline{g(\bar{z})} & z \in L \cap -\mathbb{H}. \end{cases}$$

The proof that Scharz reflection gives a holomorphic function also proves that this function is holomorphic as a function $\mathbb{C} \setminus \{z = 0\} \to \mathbb{C}$.

We can translate G into a holomorphic extension of f by conjugating $F := h^{-1} \circ G \circ h : \mathbb{C} \setminus \{p\} \to \mathbb{C}$ where p is the preimage of 1 under the continuous extension $\overline{f}$.

If f has more than one point on S^1 which is sent to 1 (i.e. more than one point on $\mathbb{R}$ is sent to ∞ under h) then we can repeat a similar procedure, except now we Schwarz reflect vertical strips bounded by the vertical lines through the points on the real line which are sent to ∞ by h.

Now notice that our extended function F may have poles. In particular, the Riemann map $h^{-1} : \mathbb{H} \to \Delta$ is given by

$$w \mapsto \frac{w - i}{w + i}$$

which has a pole whenever $w = -i$. That is to say, any $z \in \mathbb{C} \setminus \{p\}$ such that $G(z) = -i$ corresponds to a pole on the "unit disk" side. Now, recall that g mapped $\mathbb{H} \to \mathbb{H}$ since f

mapped $\Delta \to \Delta$. And so the only points which have image $-i$ under G are those in the lower half plane. That is, consider the points in the lower half plane such that $\overline{g(\bar{z})} = -i$ or the points in the upper half plane such that $g(z) = i$. Recalling the biholomorphism $\Delta \to \mathbb{H}$, these are exactly the points $z \in \Delta$ such that $f(z) = 0$. I.e. the reflected zeroes of f become poles in the extended domain. Lastly, since f has isolated zeroes (it is holomorphic), and since Δ is a compact set, f can only have finitely many zeros — any discrete set contained in a compact set is necessarily finite. And so the extended function only has finitely many poles.

Thus we have found an extended function of f which is holomorphic on all of $\mathbb{C}$ with only finitely many points removed.

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3. Let $U := \{x + iy : y \geq x^2\} \subset \mathbb{C}$, and consider ∂U its boundary in $\mathbb{C}$. Let $f : U \to \mathbb{C}$ be a holomorphic function that extends continuously to $F : U \cap \partial U \to \mathbb{C}$ with $F(\partial U) \subset \partial U$. Show that f can be extended to a holomorphic function on a strictly larger open set $V \supsetneq U$.

I think the idea with this question is that we can apply a "reflection principle" to this set with these conditions on f (similar to how we used inversion to apply a reflection principle to the unit disc in the previous question).

More precisely, the Riemann mapping theorem gives us that there's a biholomorphism from U to either the upper half plane or the unit disc. Indeed, U is simply connected and not all of $\mathbb{C}$. Moreover, since the boundary of U is "sufficiently nice", this Riemann map extends to a homeomorphism on the boundary. Then, we can define a holomorphic function on the upper half plane, by conjugating our original f with the Riemann map, and this new function satisfies the conditions for Schwarz reflection. This gives us a new holomorphic function defined on an open set strictly larger than the upper half plane. We can map this larger set back to original domain with the parabola using our Riemann map again. And then our desired extended function is given by conjugating the extended function on the "upper half plane side" by our Riemann map.

There are some details with poles which need to be thought about. We may have a nonfinite number of poles in our extension, since the parabola is not a compact set. However, they need to be isolated, and so we can remove any poles found in our extension and still be left with a strictly larger open set than the original parabola on which our new function is defined on. ∎

4. Find two bounded open subsets of $\mathbb{C}$ that are homeomorphic but not biholomorphic.

We know from the Riemann mapping theorem that any simply connected open set (which is not all of $\mathbb{C}$) is biholomorphic to the unit disc. And so, any two bounded simply connected open sets are biholomorphic to each other. Thus, we need at least one of our open sets to be non-simply connected. Moreover, since we want to find a homeoomrphism between our open sets, it follows then that both are not simply connected (and must have the same homology).

Let us conjecture that two annuli of inner radius 1 and different outer radii are not biholomorphic. Let us denote these annuli by A_1, A_2, the first having outer radius r_1 and the latter having outer radius r_2. Suppose for contradiction that there exists a biholomorphism $f : A_1 \to A_2$.

I think this is true, but I was not sure what kind of contradiction I could derive (perhaps I will come chat to you about this one next week if you have office hours.) ∎

5. Interpret the function $\log z$ on $\mathbb{H}$ as a degenerate Schwarz-Christoffel map with image a degenerate polygon.

First recall that we have a well defined logarithm on the upper half plane (take the branch cut on any ray contained in the lower half plane) and so we can write $\log z$ suggestively as

$$\log z = \int_0^z \frac{1}{\xi}\, d\xi.$$

On the other hand, writing $\log z = \log(re^{i\theta}) = \ln r + i\theta$, and remembering that points on the upper half plane have $0 < \theta < \pi$, the image $\ln(\mathbb{H})$ is the infinite horizontal strip given by $\{z \in \mathbb{C} : 0 < Im(z) < \pi\}$. This "looks" like a polygon whose corners are occuring off at infinity.

More precisely, any radial ray from $z = 0$ at some angle θ is mapped to a horizontal line ("moving to the right") at $im(z) = \theta$. Consider the image of the boundary rays under $\log z$. We have that the ray $(0, \infty)$ is mapped to the "lower" boundary component $im(z) = 0$. On the other hand, the ray $(0, -\infty)$ is mapped to the "upper" boundary component at $im(z) = \pi$.

If we want to think of this map as a degenerate Scharz-Christoffel map, then we might want to think of what a "continuous extension" of $\log z$ to the boundary might mean (of course, this is a heuristic argument since $\log z$ is ill defined at $z = 0$). Approaching $z = 0$ from $+\infty$ gives a point "infinitely far to the left" on the line in the image $im(z) = 0$ whereas approaching $z = 0$ from $-\infty$ gives a point also "infinitely far to the left" but now on the line $im(z) = \pi$. If we want to think about a continuous extension, we can then think of these two boundary lines as meeting at infinity. Then we can think of this infinite strip as having a "corner" at $z = \infty$ in the image, or at $z = 0$ on the boundary of the upper half plane. And indeed, this is manifest in $\log z = \int_0^z 1/\xi d\xi$ since the corners in Schwarz-Christoffel maps are supposed to occur at the images of the denominator points. That is all to say, we can think of the image of $\log z$ applied to the upper half plane as a degenerate polygon, with one corner at $z = \infty$. If we want to extend our source domain to include the point at infinity, then we can also think of the image of $\log(z)$ as having another corner "infinitely far to the right", or at the image of ∞.

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